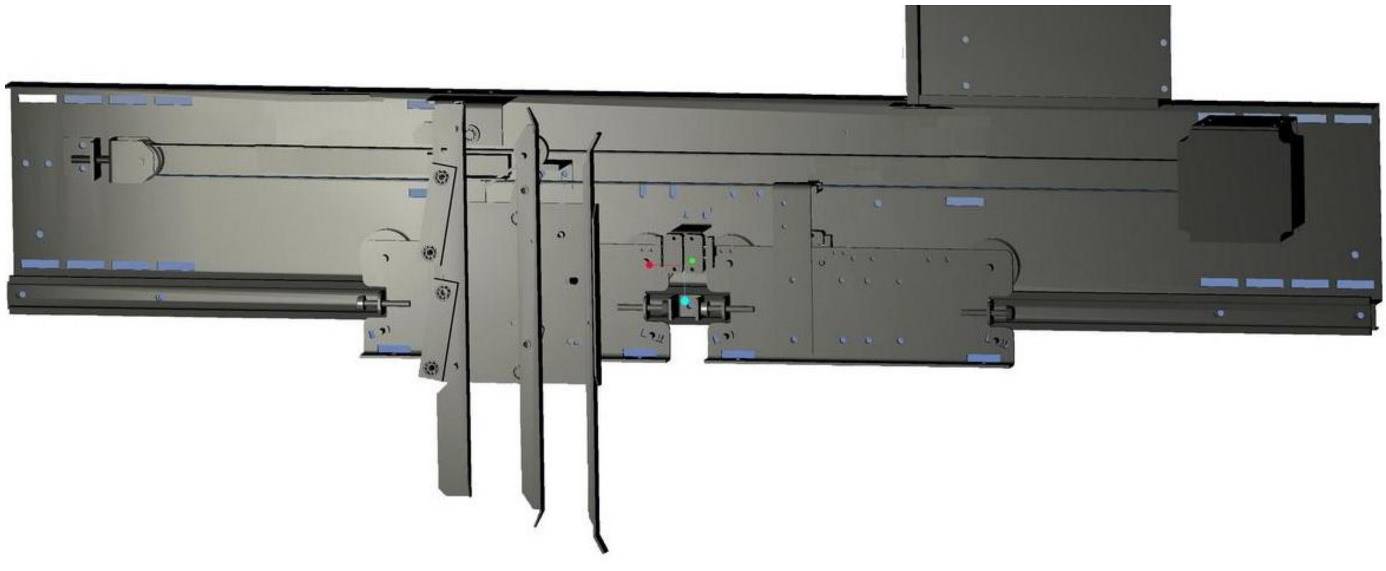
DO3000
Center-opening Permanent Magnetic VVVF Door Operator

Instructions Operation



Operation Instruction

Safety Notes

In this manual, the safety notes are divided into the following two categories:

⚠DANGER: DANGER indicates a hazardous situation which is caused by operation not following the instructions. It may result in serious injury or even death.

⚠CAUTION: CAUTION indicates a hazardous situation which is caused by operation not following the instructions. It may result in medium or slight injuries, as well as the damages to the equipment.

During installation, commission and maintenance of this system, please read this chapter carefully, and operate in compliance to the safety notes specified in this chapter. We will not be responsible for any damage or loss caused by operations against regulation.

1 Installation

⚠DANGER
■ Please install the equipment in the fire-retardant materials (like metal) to prevent fire! ■ Keep it away from flammable material to prevent fire!

⚠CAUTION
■ Prevent the lead head or screw from dropping into it to avoid causing damage to the controller! ■ Install the controller into the place where there is little libration and no direct sunlight. ■ Install the equipment into the weight-durable places to prevent injuries caused by the dropping of the equipment. ■ Do not make the installation if the controller is found broken during unpacking! ■ Do not make the installation if the shipping list is inconsistent with the goods! ■ Handle with care; otherwise it may cause damage to the equipment! ■ Do not touch the elements of the controller with your hands; otherwise it may cause the electrostatic damage!

2 Wiring

⚠DANGER
■ The instruction of this manual must be obeyed. The constructions must be carried out by the professional electrical engineering staffs to prevent from getting an electric shock or injury! ■ There must be a breaker between the controller and power supply, or it may cause fire! ■ Please carry out the correct standard earthing of the controller following the specification; otherwise there may be the risk of getting an electric shock!

⚠CAUTION
■ The input power supply must not be connected to the output terminals (U, V, W) of the controller. Pay attention to the marker of the wiring terminals, and do not make the wrong wiring! Otherwise it may cause the damage to the controller! Make sure that the wiring configuration is complied with EMC requirements and the local safety standards. Otherwise it may cause accidents! ■ The communication line must be STP, the lay of which is 20~30 mm, and the shielding layer must be grounded. ■ Make sure that the nominal voltage of the product is the same with that of the alternating current power supply so as to avoid the injury and fire! ■ Pay attention to check if there is short in the peripheral circuit connected to the controller and whether the connected circuit is fastened. Otherwise it may cause damage to the controller! ■ No part of the controller should take withstand test, as the product has taken it in the factory. Otherwise it may cause accidents!

3 Power-on

⚠DANGER
■ The controller can only be powered on after the cover board be put in place. Do not open the cover board after power-on. Never touch any input and output terminal of the controller, or you may get an electric shock! ■ All the wirings of the peripheral fittings must be complied with the instruction of this manual, and be carried out correctly following the circuit connection method provided in this manual. Otherwise it may cause accidents! ■ Never change the controller manufactory parameter at will. Or it may cause damage to the equipment! ■ Non-professional technician shall be forbidden to detect the signal during the operation. Otherwise it may cause damage to the people or the equipment!

4 Maintenance, inspection and replacement of components

⚠DANGER
■ Please do not repair or maintain the equipment lively. Or you may get an electric shock! ■ Personnel without professional training shall not be allowed to repair or maintain the controller. Otherwise it may cause damage to the people or the equipment! ■ The parameter setting must be done after changing the controller; all of the plug-pull plug-ins must be plugged and pulled with the power off. ■ The input power needs to be cut off during the maintenance and inspection, do it 5 minutes later to avoid getting an electric shock.

5 Usage beyond nominal voltage

If the external voltage is not within the allowable working voltage range specified in the manual, using the controller may cause damage to its parts easily. If it is required, please make a transformation using the corresponding voltage-boosting or voltage-reducing devices.

6 Protection against lightning impulse

This series of controllers are installed with lightning over-current protection device and can protect themselves from induction lightning to some extent. For the places where the lightning presents usually, the customer should add protection device to the front end of the controller.

7 Height above sea level and derating

At the place where the height above sea level is above 1000 meters, the heat dissipation effect of the controller may be reduced by the rareness of air. Then it will be necessary to make the derating. For the detail please make technique consultation with our company.

8 Notes on disusing controller

The electrolytic capacitor of the main circuit and the print plate may blow up when they are set on fire. The plastic parts will produce toxic gas when they are set on fire. Please treat them as industrial refuse.

9 About applicable motor

This controller is applicable for permanent magnetic synchronous motor; please select the controller as per the name plate of the motor.

In order to reach better control, it's required to make motor orientation in accordance with the actual motor situation.

The short of cable or the interior of the motor will cause the controller alarm even damage. For this reason, please take insulation and short test to the motor and cable which are installed for the first time, and this test is needed in the daily maintenance as well. Notice, the controller and the parts to be tested must be separated thoroughly during this kind of tests.

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1 Product introduction

DO 3000 center-opening permanent magnetic VVVF door operator is a doors driving system, which adopts permanent-magnet synchronous motor drive and stepless speed regulation and frequency conversion control. The product is highly efficient, reliable and easy to operate. It can reach the optimum door opening/closing speed curve with low mechanical shock. DO 3000 center-opening permanent magnetic VVVF door operator is applicable for a door system with a 700-1200mm net width of door opening.

Its main features are as follow:

1 (stationary) autotuning of synchronous motor rotational angles

2 auto tuning of door width

3 automatic opening/closing door demonstrations

4 fault alarm and automatic protection function

5 optional baffle detection function

6 synchronous and asynchronous door cutter selection functions

1.1 Technical parameters

Input voltage: AC220V±20%;

b) Transducer:

Rated frequency: 50/60Hz;

Output voltage: 0～220V;

Output frequency: 0～50Hz

c) Motor:

Rated voltage: AC100/125V;

Rated rotate speed: 180r/min;

Rated power: 43/94W

1.2 Operating conditions

a) Altitude 1000 m max., 100% of rated current output;

1000 to 2000 m, 95% of rated current output;

2000~2000~3000 m, 85% of rated current output;

b) The highest relative humidity of the dampest month in the operation site is 95%, while the average lowest temperature of the same month is not higher than 25 ℃;

c) The power supply voltage is limited within the range of ±15% of the rated voltage;

d) There shall be no corrosive and flammable gas or conductive dusts in the surrounding air;

2 Installation instruction for door operators

2.1 General illustration for door operator installation

General illustration of door operator installation: see Figure 2-1

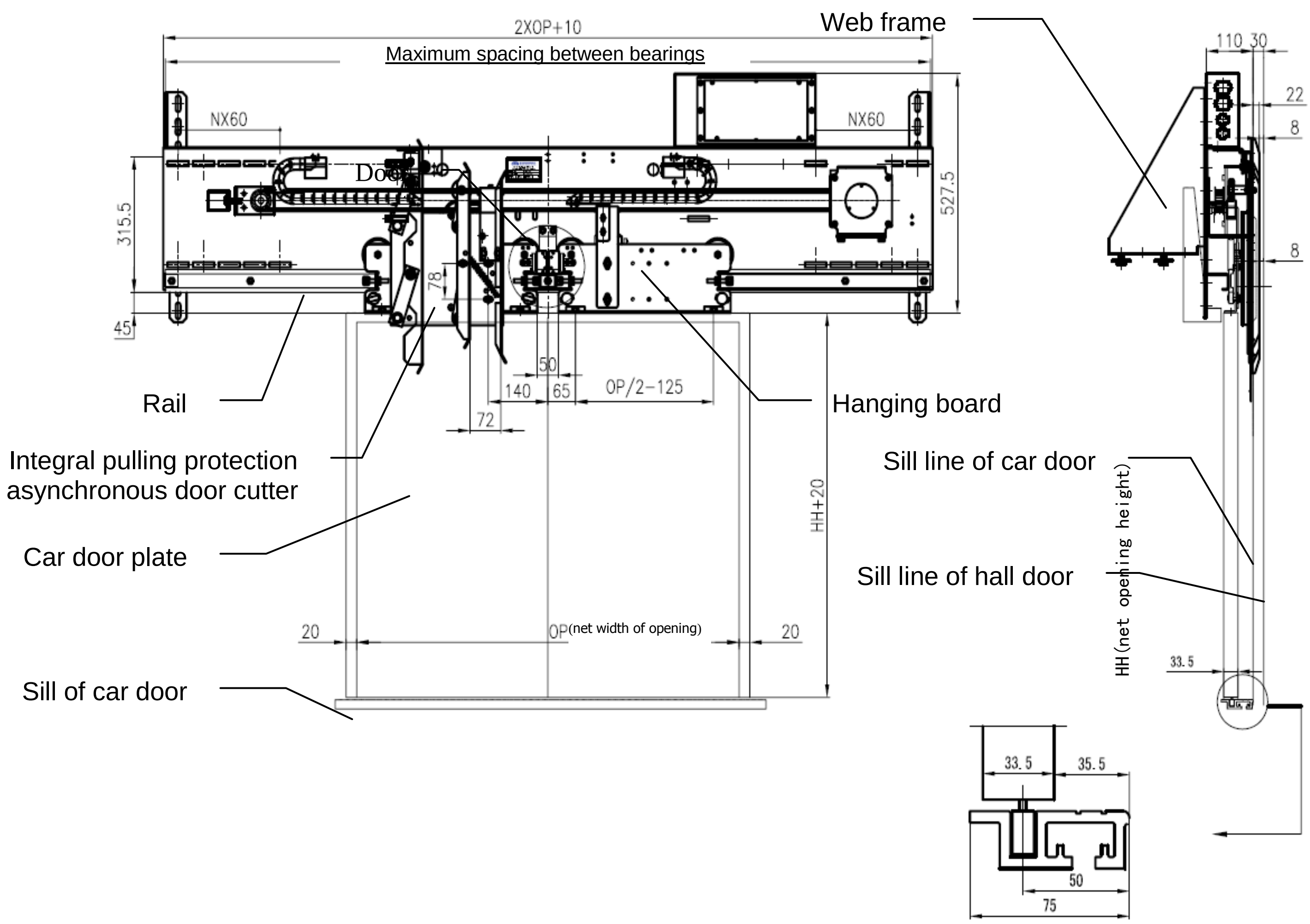


Figure 2-1 General illustration of door operator installation

2.2 Installation and adjustment of door operator

2.2.1 Installation of door operator mounting bracket

2.2.1.1 Installation of car roof

Fix the car roof mounting bracket of door operator on the C type slot on the car roof with fixing bolts. See Fig. 2-2

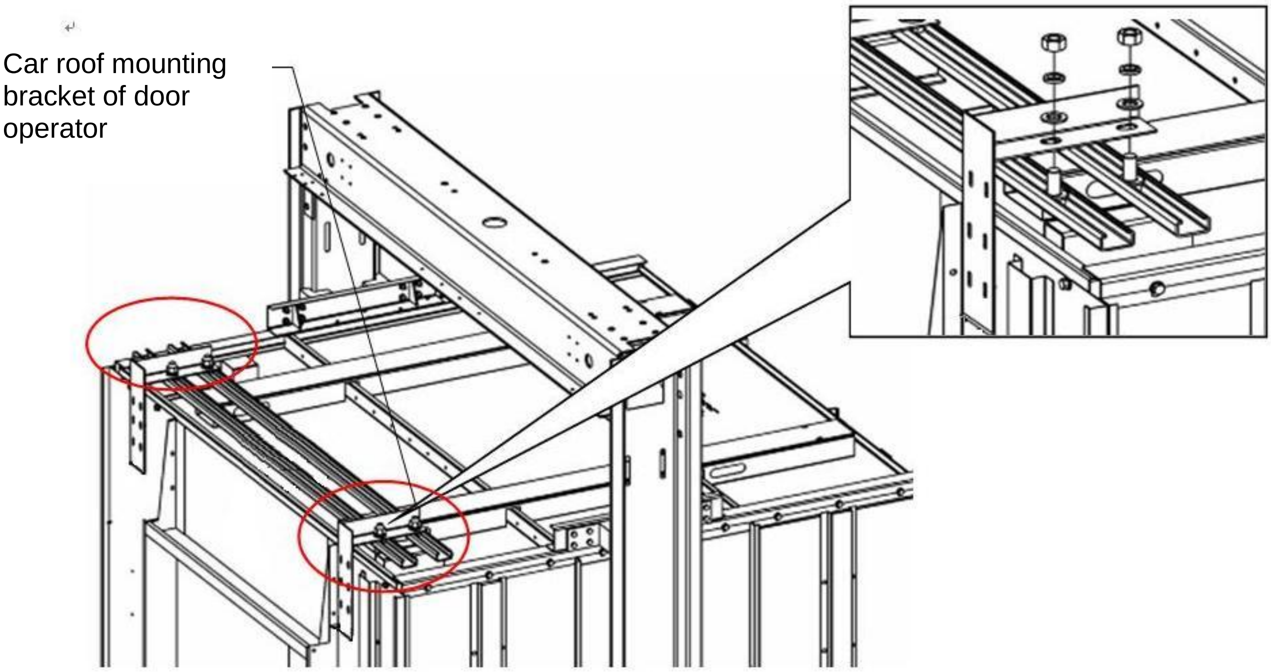


Fig. 2-2 Installation of mounting bracket of door operator (installation of the car roof)

2.2.2 Installation and adjustment of DO 3000 door operator

The connecting bolts should be mounted into the mounting holes on the door head of the door operator in advance. Then connect the door operator and its mounting bracket with connecting bolts.Adjust the position of the door operator after installation. It is required that the door operator rail should be parallel to the car door sill. Also, the door operator should be kept vertical by checking with plumbline. Splines can be used to keep parallelism. The bias of parallelism and verticality should be limited to less than 1 mm.Adjust simultaneously both the door operator center and exit and entrance center to make them aligned; Adjust the height of the door operator to make sure it is suitable for installation of door plate.See Fig. 2-3 Installation illustration of car roof of door operator

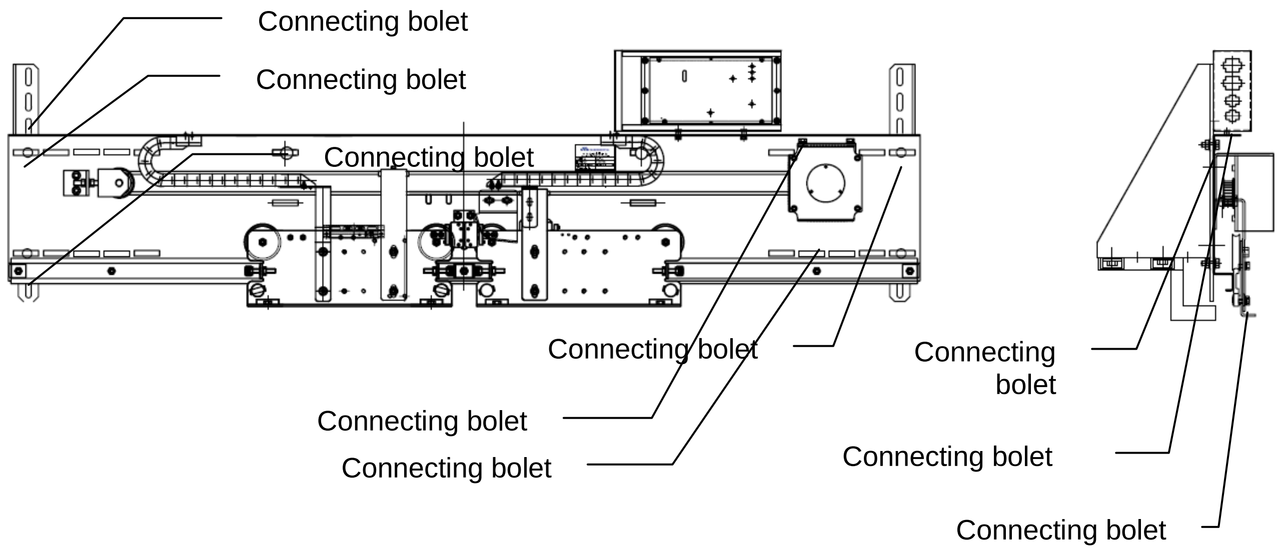


Fig. 2-3 Installation illustration of car roof of door operator

2.2.3 Installation and adjustment of the door cutter

2.2.3.1 Installation of the pulling protection asynchronous door cutter

The pulling protection asynchronous door cutter is installed with fixing bolts on the hanging board of the door operator, see Fig. 2-4, 2-5.

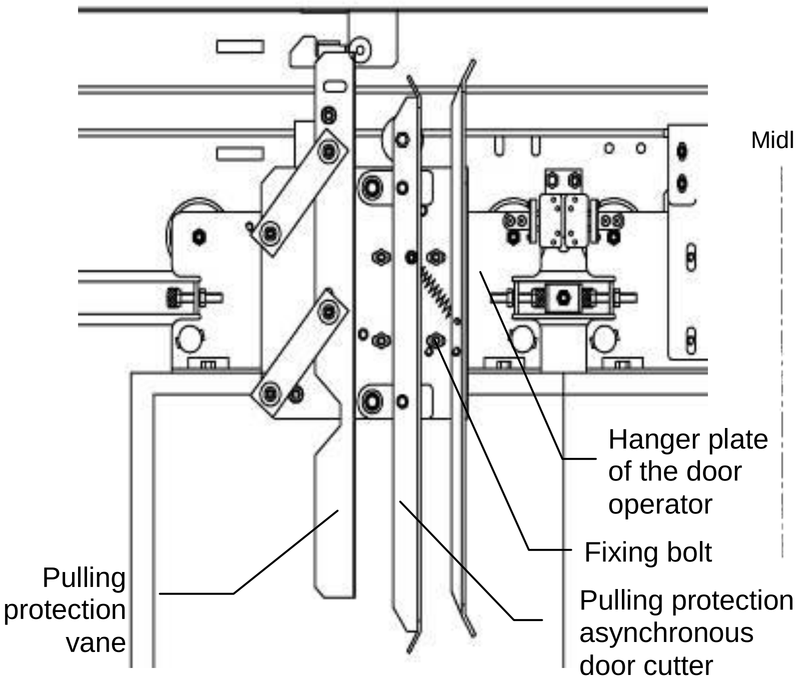


Fig. 2-4 Configuration of the pulling protection vane

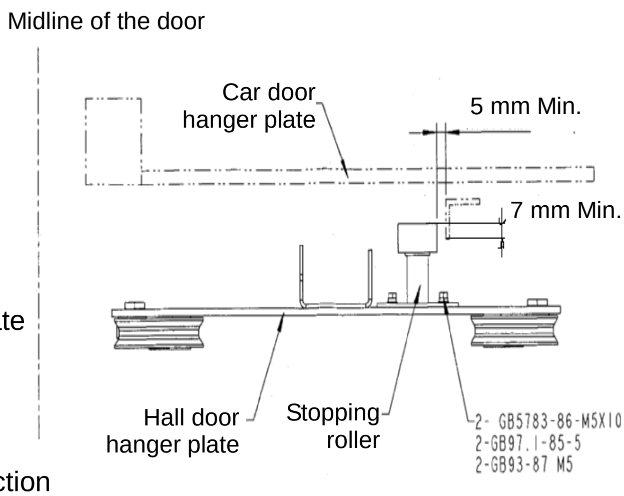


Fig. 2-5 Installation of the pulling protection asynchronous door cutter

Installation steps:

1. Adjust the back bag surface of the hanger plate to the vertical condition.
2. Open the door manually, and keep the door clearance at 200 mm, the roller on the movable blade

being kept away from the door-lifting block of the door cutter. Fix the main part of the door cutter to the hanger plate back bag using fixing bolt, so as to make the distance from the fixed blade and the lock matching surface to the door center reaches 111 mm after the door reached closed limit.

3. Make sure that the locking screw of the pulling protection vane is locked, and then close the door operator in place. Adjust the front and rear locations of the pulling protection accessories to make a good match between the rubber roller on the accessories and the hook fixed on the pulling protection vane. Then adjust the left and right location of the theft-against accessories to make a perfect touch and match between the rubber roller on the accessories and the hook hem of the pulling protection vane. Fasten the pulling protection accessories.

Requirements for installation (notices):

1. The clearance between the door cutter and the hall door sill is 5 to 10 mm (allowing for the clearance between the door cutter and the upper sill casing).

2. The occlusion length between the door cutter and hall door lock roller is 7 to 9 mm. In leveling position, the door lock roller is in the middle of the two blades of the door cutter so that the roller will not collide with the cutter blade while the elevator is running.

3 The verticality of door cutter is limited within 0.5 mm. The space between two blades is 70 to 72 mm when the car door is closed completely.

4. The door cutter is not allowed to collide with the door lifting block, and the position of the door cutter roller and the door lifting block should be suitable.

5. When the door reaches closed limit, the relative position of the stopping roller (installed on the hall door hanging board) to the pulling protection vane is shown in Figure 2-19. In addition, the distance between the stopping roller and the vane should be longer than 5 mm when the vane is in uplift position (during the door closing process, the roller on the vane will rise under the effect of the bracket of the door operator).In this case, the bracket of the vane will separate from the bracket of the door operator.

6. When the elevator is in leveling position, if the car door is pulled open with hand, the vane will start to deflect downwards with the two swing links to the central position of the door and the stopping roller will start to contact with the vane.

7. When the elevator is not in leveling position, if the car door is pulled open with hands, the vane will start to deflect downwards with the two swing links to the central position of the door. In this case, if the door board is kept in open position by hand, the brackets on the vane and on the door operator will clasp each other so that the door can not be opened.

2.2.4 Adjustment of hanging board resistance

When there is much resistance of operation, adjust the clearance of the lower roller of the hanging board and the rail to 0.1~0.3 mm as shown in Fig. 2-6 to make the door open and close smoothly.

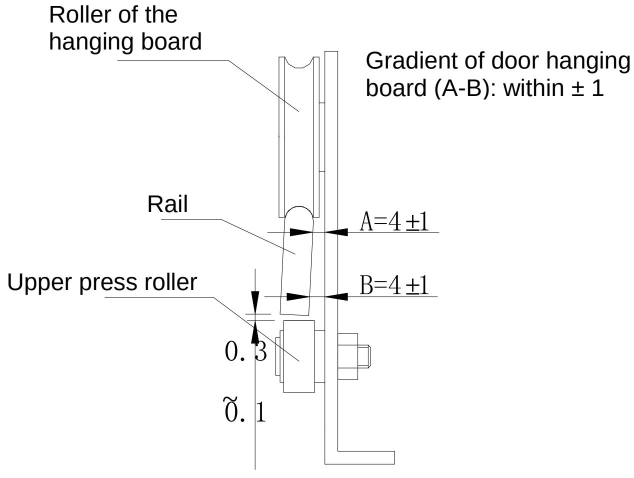


Fig. 2-6 Adjustment of door hanging board

2.2.5 Wiring of door protection

2.2.5.1 Wiring of safety edge

When the safety edge is equipped, make the Wiring of the safety edge as Fig. 2-7 shows. Fasten the cable with inversive bandage on the car door, and connect it through the drag chain bracket as well as the drag chain to the control circuit.

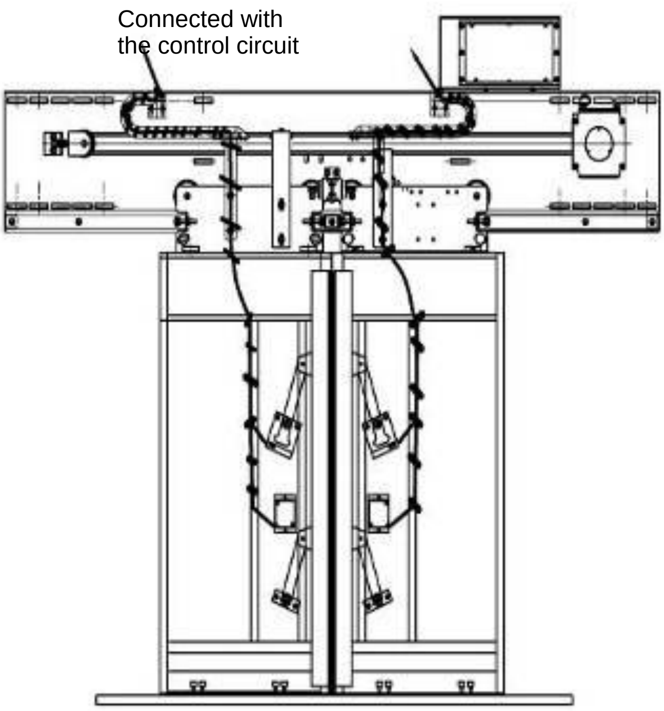


Fig. 2-7 Wiring of safety edge (with drag chain)

2.2.5.2 Wiring of light curtain

When the light is equipped, make the wiring of light curtain as Fig. 2-8 shows. Fasten the cable with inversive bandage on the car door, and connect it through the drag chain bracket as well as the drag chain to the control circuit.

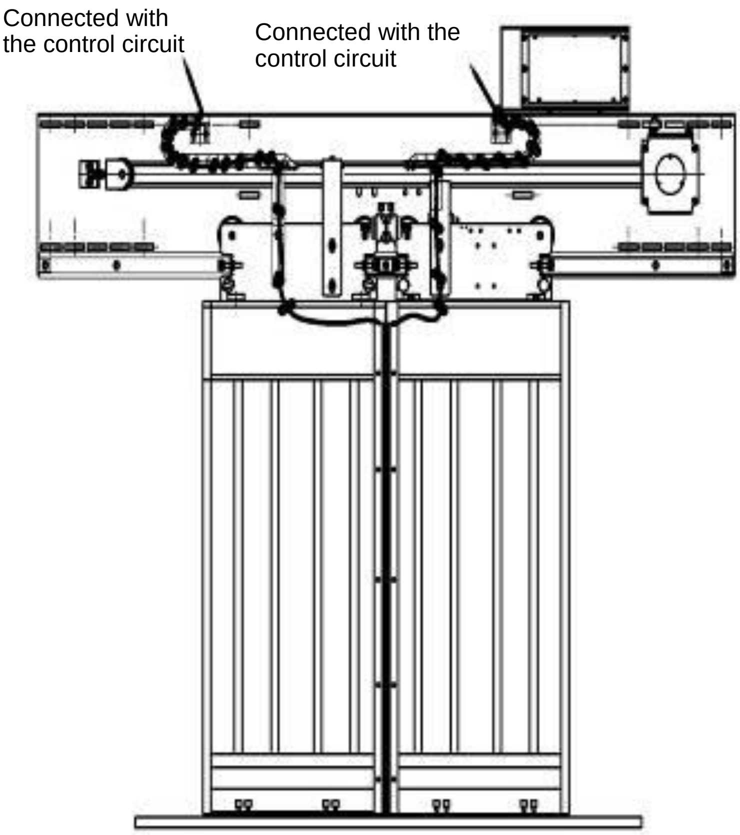


Fig. 2-8 Wiring of light curtain (with drag chain)

Requirements for light curtain installation:

1. The surface at the bottom of the light curtain should be flush with that of the fixing bracket, and the distance between the surface at the bottom of the light curtain and the surface at the bottom of the car door board should be 10 mm at least; The two light curtains should be 10 to 20 mm apart from each other when the door is closed; All the mounting holes of the light curtain should be fixed on the door leaves.

2. The light curtain line should be fixed securely along the strengthening rib of the door board with tape in reverse dragging fashion, and then it is connected to the junction box of light curtain on the car roof.

3. After the light curtain is installed, it should be grounded. The ground line is connected to the car door with bolts and is made into a loop connected with the ground line on the car roof.

3 Electrical adjustment

3.1 Wiring of transducer

See the wiring in Fig. 3-1.

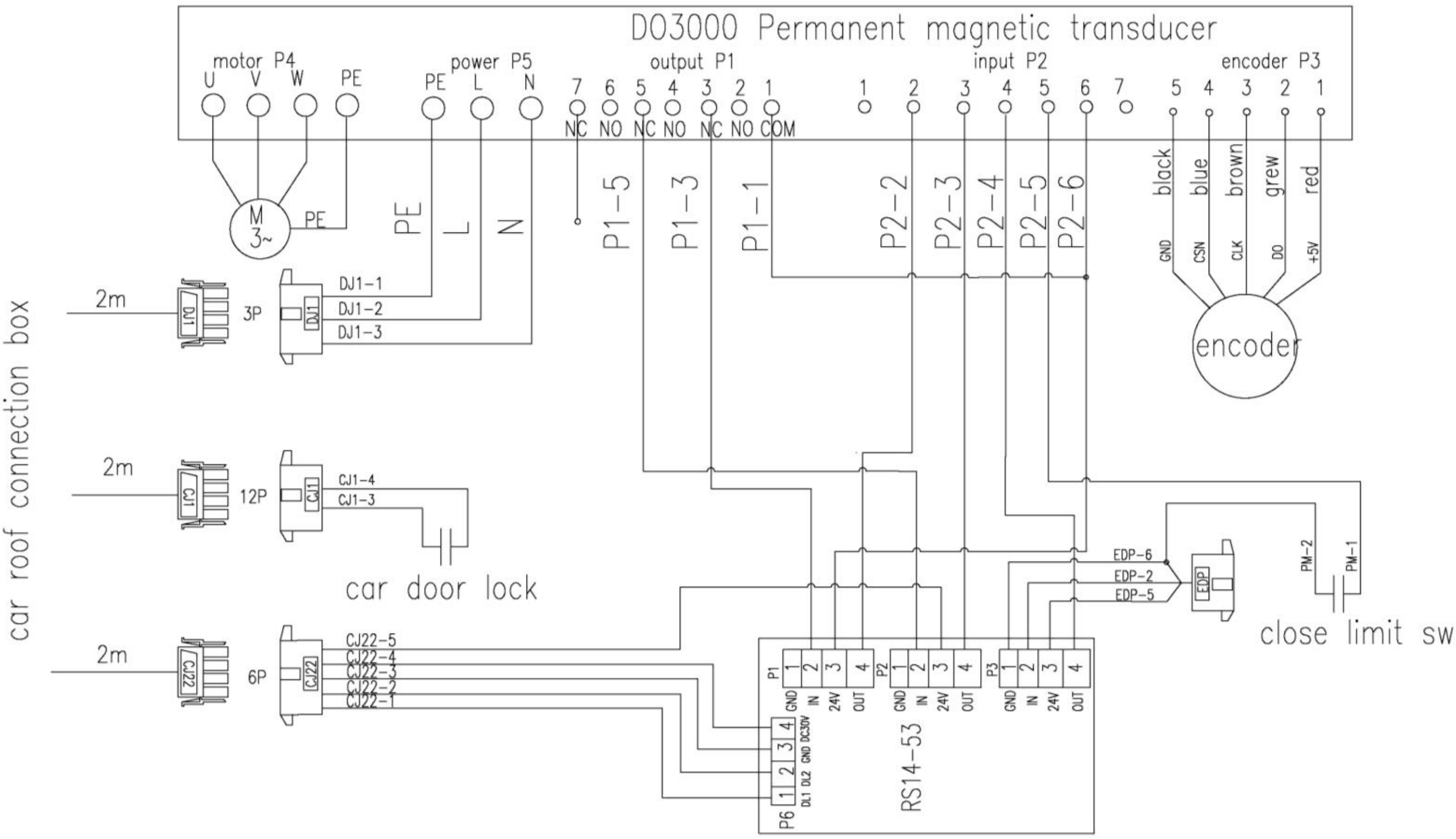


Fig. 3-1 Transducer wiring diagram

See the wiring of the light curtain and safety edge in Fig. 3-2

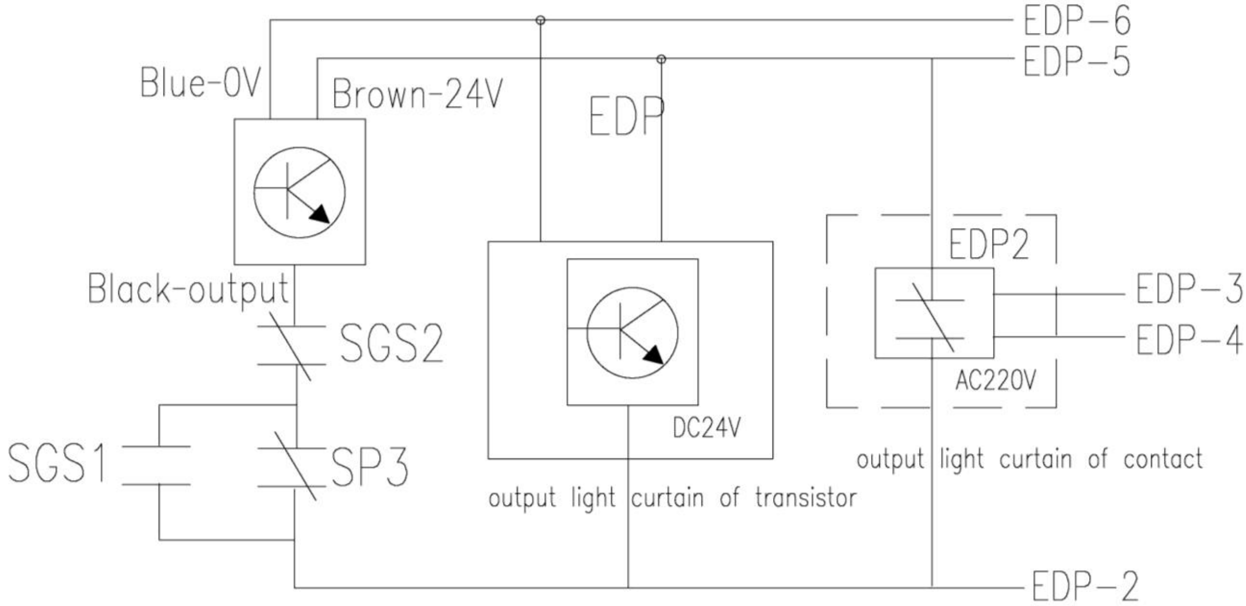


Fig. 3-2 Wiring diagram of light curtain and safety edge

3.2 Controller interface

3.2.1 Casing of controller

See the casing of controller in Fig. 3-3.

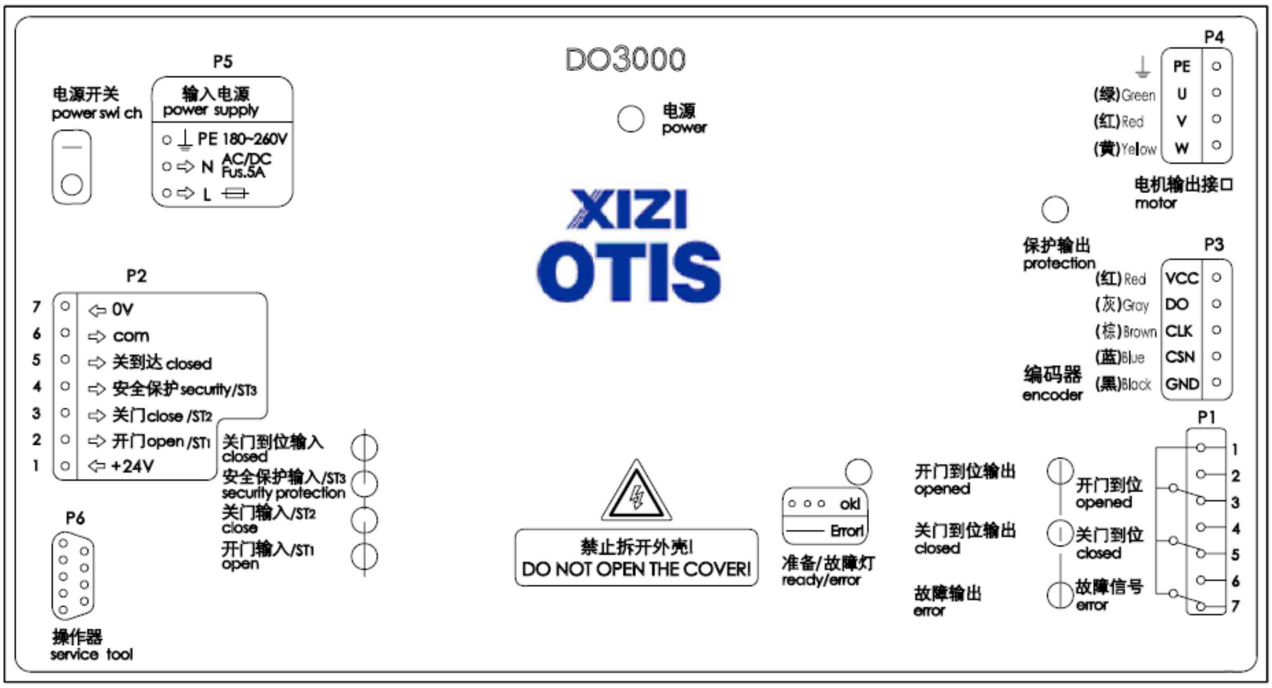


Fig. 3-3 Casing of controller

- Note: Those solid circles in the figure represent indicators for corresponding functions.

  Power: Constantly on when the power is normal

  Preparation/ fault light: Glint when it is normal, constantly on when fault occurs.

  Door closed limit input: When the door reaches closed limit, the limit switches signal inputs, constantly on as there is signal.

  ST 1/ST 2/ST 3: Signal of ST code inputs, constantly on as there is signal.

  Door opened limit output: When the door reaches opened limit, the transducer outputs the signal of reaching opened position, constantly on as there is signal.

  Door closed limit output: When the door reaches closed limit, the transducer outputs the signal of door closed limit, constantly on as there is signal.

  Fault output: As fault occurs, the transducer outputs the signal of fault, constantly on as there is signal.

  Protective output: The light is off when there is no motor output, and constantly on when there is motor output.

3.2.2 Definition and description of input and output ports

See the definition and description of the input and output ports separately in Tab. 3-1 and 3-2.

Tab. 3-1 Definition and description of the output port

Terminal	Description
P1-1	Common terminal of the output relay
P1-2	NO contact output of the opened limit
P1-3	NC contact output of the opened limit
P1-4	NO contact output of the closed limit
P1-5	NC contact output of the closed limit
P1-6	NO contact output of the fault signal
P1-7	NC contact output of the fault signal

Tab.3-2 Definition and description of the input port

Terminal	Description
P2-1	24 V
P2-2	Opening signal/ST 1
P2-3	Closing signal/ST 2
P2-4	Safety actuated signal/ST 3
P2-5	Closing door reached magnetic switch signal
P2-6	COM
P2-7	OV

3.3 Application of server

3. 3. 1 Introduction of server application

The private server with easy operation and high efficiency is adopted. See the server in Fig. 3-4.

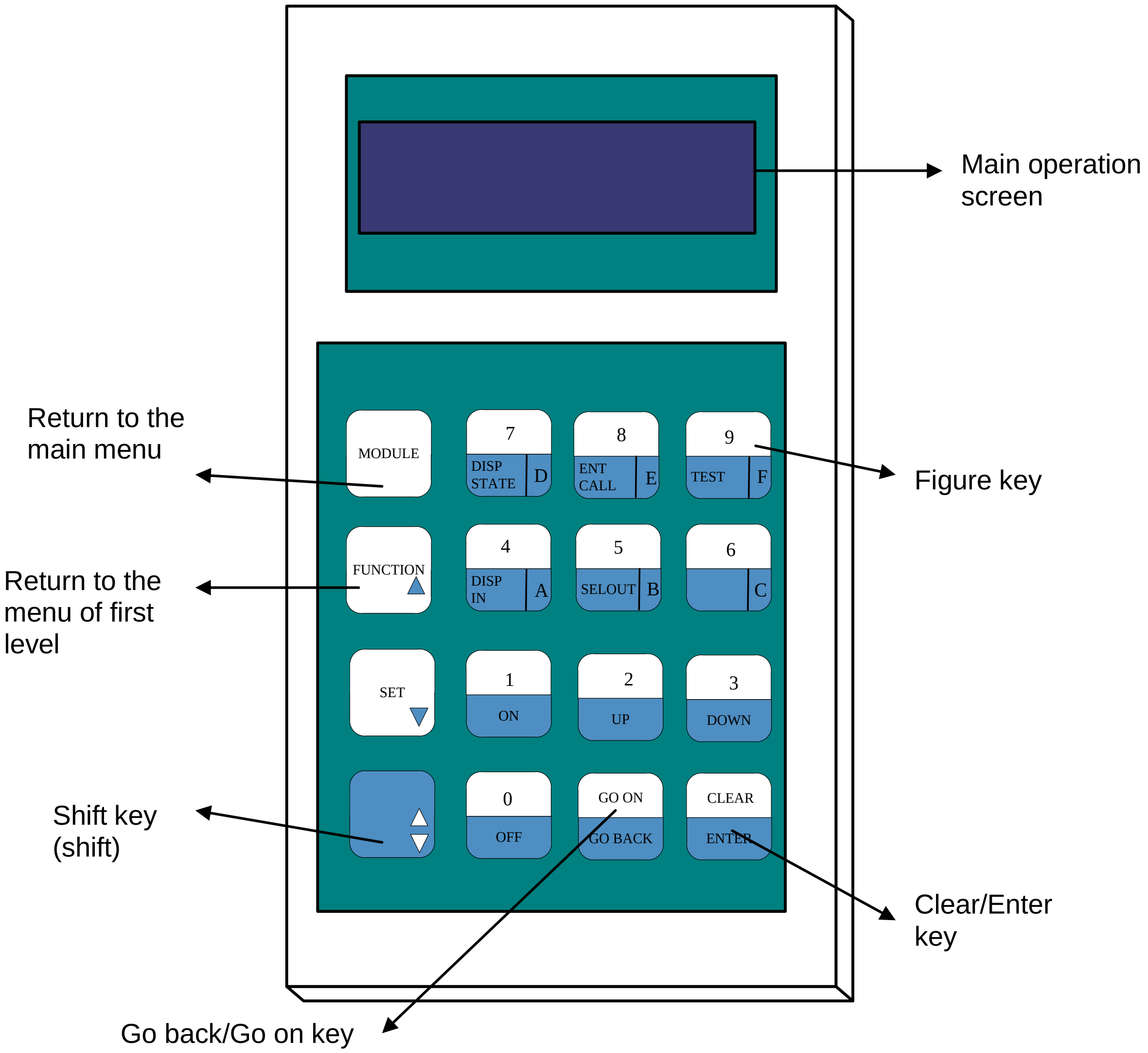
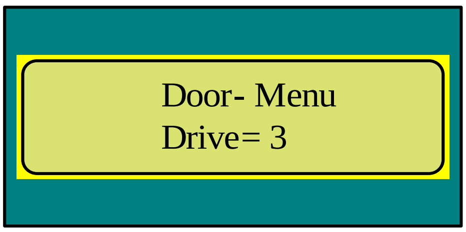


Fig. 3-4 Server

The three ranges of key-press on the right of the server are divided into upper and lower rows. If you want to use the upper functions, press the key directly; if you want to use the lower functions, press Shift key + this key. You should pay great attention to the GO ON, GO BACK, CLEAR and ENTRER described in the following adjustment document.

3.3.2 Operation examples

The operation screen of the server shows as following as power is on:

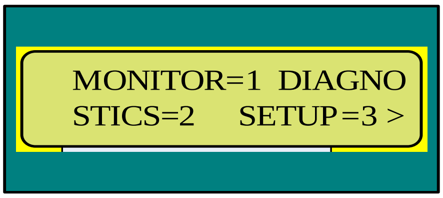


This interface is

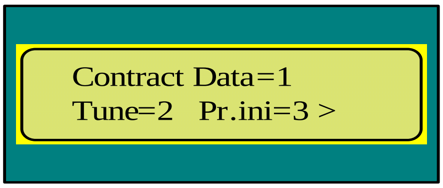
the main interface of the server. Press MODULE on the left upper end of the keyboard, and then return to this interface directly.

For example: enter menu 3311 (curve parameter)

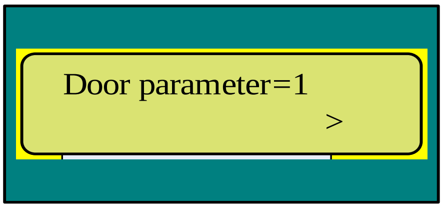
Press 3 DOWN at the interface of the main menu, and then enter the following interface



Press 3 DOWN again to enter the following interface

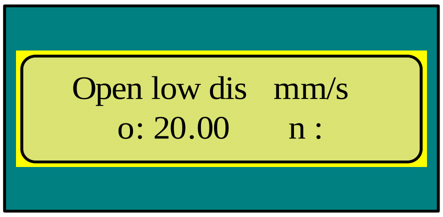


Press 1 ON again to enter the following interface



3.3.3 Data modification

The server operation interface can show two rows of data. The original data (old, simplified as o) show on the left of the second row, and the new data (new, simplified as n) can be input from the right side as shown in the following interface.



The method of entering data: for example, to enter 40, first press the figure key "4", and then press "0" for twice, and then you can see 40.00 on the right. After affirming that this is the data required to be entered, press ▲▼ + CLEAR ENTER .If you press 400.00 carelessly, press CLEAR ENTER , and then one "0" will be cleared.

3.4 Adjustment description

3.4.1 Please make sure of the following prior to the door operator adjustment:

- The setting of the system address are as following:
  DOL (door opened limit): 53-1
  DOL (door closed limit): 53-2
  EDP (light curtain): 53-3
- There is pulling protection vane bolt on the soleplate of the door cutter, and the pulling protection vane is lifted as is shown in Fig. 3-5

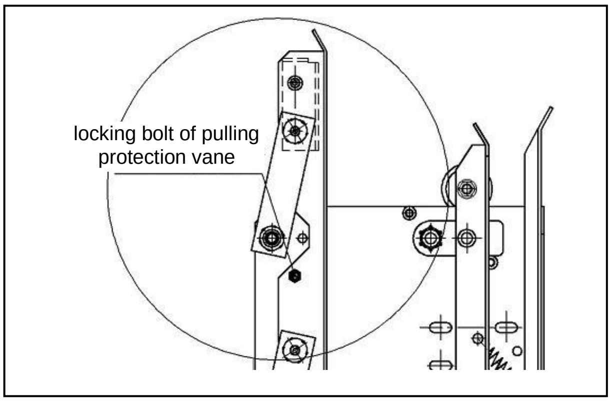


Fig. 3-5 Pulling protection vane bolt

- The elevator is being maintained.
- The car roof emergency press button is pressed.
- Cable CJ 1, DJ 1, CJ 17 and CJ 22 is attached to the door operator, connect them correctly as shown in Fig. 3-1.
- When the installation of the door operator is completed and the wiring is made correctly, before the power is on, push and pull the door by hand, you may find that there is resistance of door operation. If you pull off the three-phrase power plug, then the door will operate smoothly.

Note: as the transducer has the function of power-off and anti-collision, pulling and pushing the door will result in certain resistance before the operation.

The following operation must be carried out strictly in accordance with the elevator safety specification!

- ※ The match of the transducer and motor has finished when the complete machine is produced by the factory. Operate following the basic adjustment steps in 3.4.2.
- ※ If the transducer or motor is replaced, operate according to the senior adjustment steps in 3.4.3.

3.4.2 Basic adjustment steps

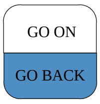
3.4.2 1 Power-on of door operator

Switch on the transducer switch; if the power indicator light of the transducer is on, the power is normal.

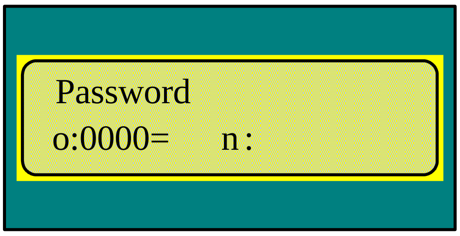
3.4.2.2 Enter password

Enter menu 3311, turn to Password parameter, enter the password 8888, and then enter the password 4321.

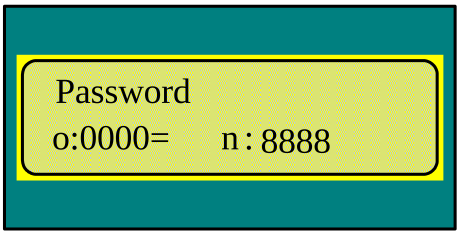
The detailed operations are as following:

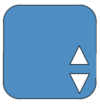
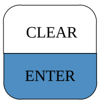
Enter menu M 3311, press 

to turn to the last parameter:

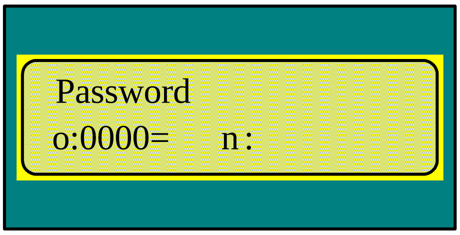


Enter the password 8888:

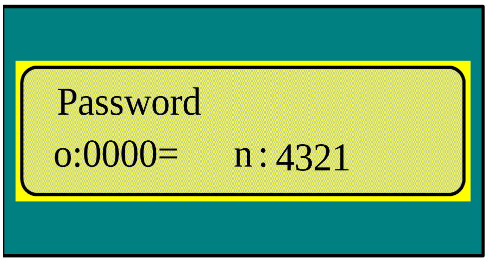


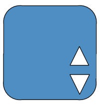
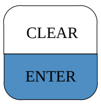
Confirm, then press  + 

.Now the menu interface changes into:

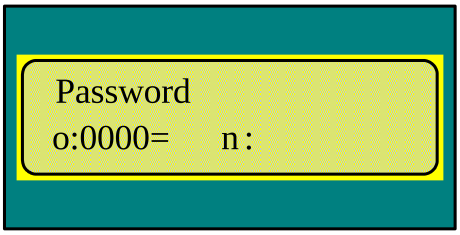


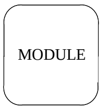
Enter the password 4321:



Press  + 

to confirm. Now the menu interface changes into:



Press 

to return to the main menu.

If you do not operate the server, it's necessary to press the password again.

3.4.2.3 Autotuning

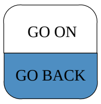
Step 1, enter menu 3311, and change the operation mode Run command source into 0.

Step 2, enter menu 3312, and change Feedback mode into 0.

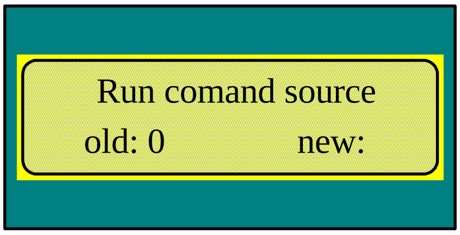
Step 3, enter menu 334, and press ENTER, then begin the autotuning.

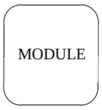
Note: the above steps must be followed step by step. If you begin the autotuning directly without step 1 and 2, the door operator may operate abnormally. It won't return to the normal unless the power is off (until the power light is off completely) and on again.

The detailed operations are as following:

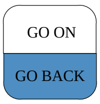
Enter menu M 3311, then press 

and select the following parameter. If it is not 0, set it as 0:

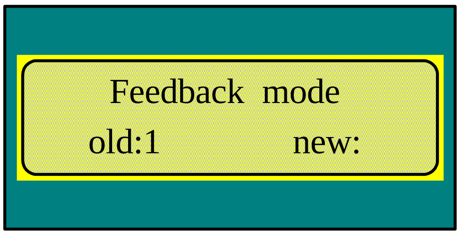


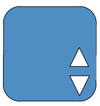
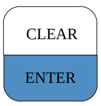
Press 

to return。

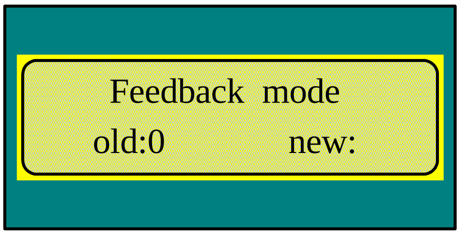
Enter menu M 3312, and press 

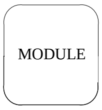
to choose the following parameter:



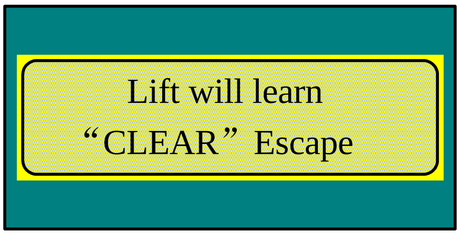
If the value is not 0, set it as 0 and press  + 

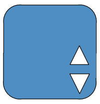
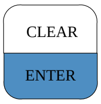
to confirm.



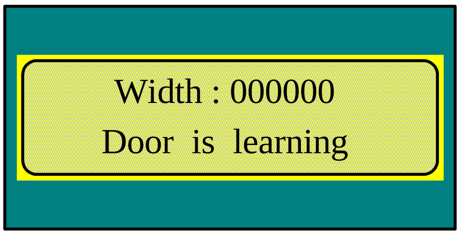
Press 

to return to the main menu, and enter menu M 334：



Press  + 

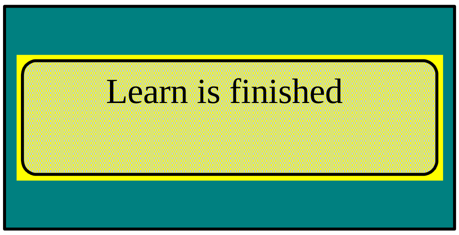
to begin the autotuning. The beginning interface of autotuning is as following:

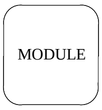


The operation process of the door during the autotuning is:

Closing the door → door closed limit → opening the door → door opened limit → closing the door → door closed limit

The finished interface of the autotuning is as following:



Press 

to return to the main menu.

After completing the autotuning (the last door reaches closed limit), the motor is disabled.

After completing the autotuning, enter menu M3311 and the value of "DR half range" can be checked. The value of center-opening door operator should be about half of the opening width plus 25 mm, and that of the sliding door operator should be about opening width ＋ 50 mm.

3.4.2.4 Demonstration operation of door operator

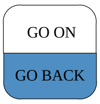
1, enter menu 3312, and check whether the Feedback parameter is 0. If it's not, change it into 0.

2, enter menu 3311, and change the operation mode Run command source into 2.

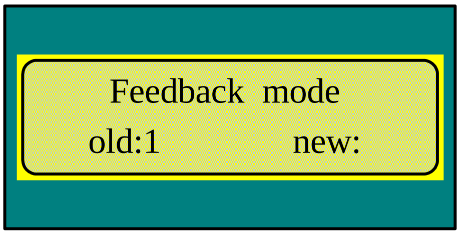
3, enter menu 313, and press ENTER to make the demonstration operation.

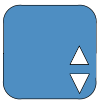
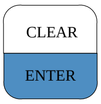
4, enter menu 313, and turn to Stop, press ENTER, then press ENTER to stop the demonstration operation.

The detailed operations are as following:

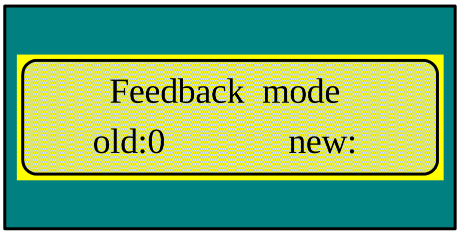
Enter menu M 3312, and press 

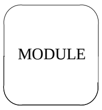
to choose the following parameter:

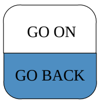


If the value is not 0, set it as 0 and press  + 

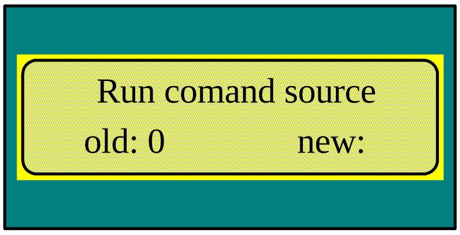
to confirm.

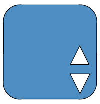
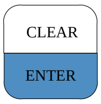


Press 

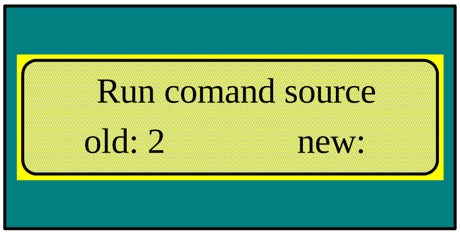
to return to the main menu, enter menu M 3311, and press 

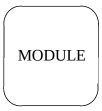
to choose the following parameter:



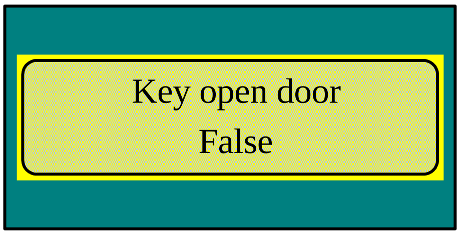
If the value is 0, set it as 2 (if it's not 0, set it as 0 first), press  + 

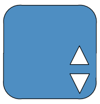
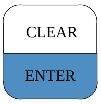
to confirm.



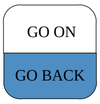
Press 

to return to the main menu, and enter menu M 313. The interface is as following:

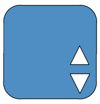
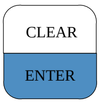


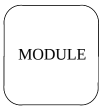
Press  + 

to begin the demonstration operation. The door operator opens and closes the door continuously.

When you need to stop, press 

, and select the following parameter

Press  + 

to stop the demonstration operation, press 

to return to the main menu。

3.4.2.5 Normal condition setting of door operator

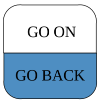
Before set the normal condition of the door operator, the emergency button must be reset, and the car-top must be in maintenance. Observe the ST code indicator light ST 1, ST 2 and ST 3 on the transducer panel; ST 3 should be the only one that is on.

1, enter menu 3311, and change the value of Run command source into 0.

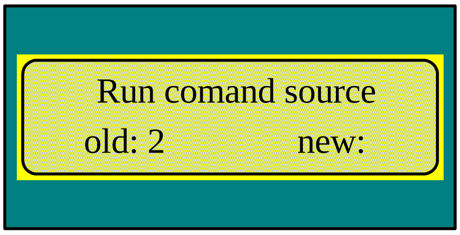
2, enter menu 3312, and change Feedback mode into 1.

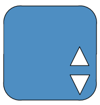
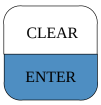
3, enter menu 3311, and change the value of Run command source into 5.

The detailed operations steps:

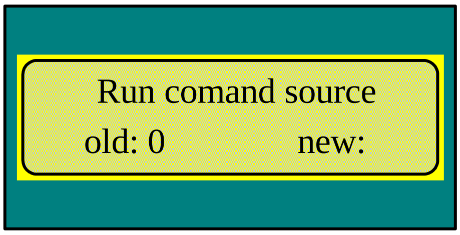
Enter menu M 3311, and press 

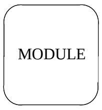
to choose the following parameter:

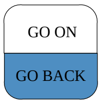


If the value is not 0, set it as 0 first, and press  + 

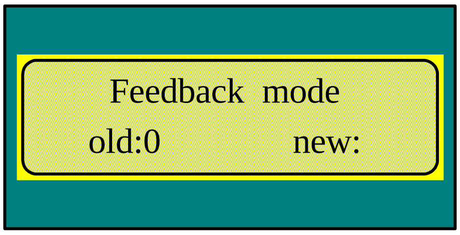
to confirm.

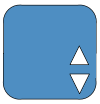
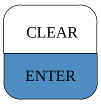


Press 

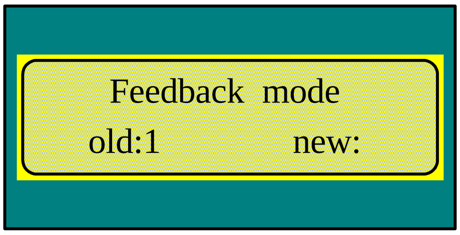
to return to the main menu, enter menu M 3312, and press 

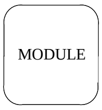
to choose the following parameter:

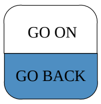


Set the value as 1, press  + 

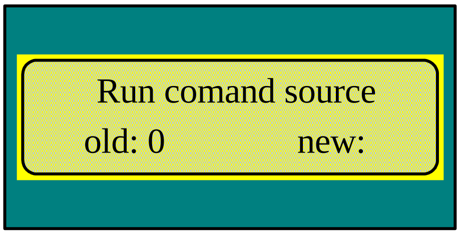
to confirm.

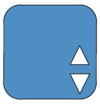
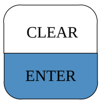


Press 

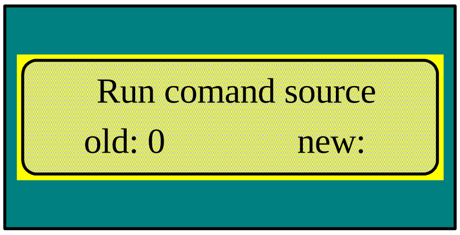
to return to the main menu, enter menu M 3311, and press 

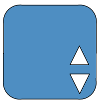
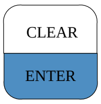
to choose the following parameter:



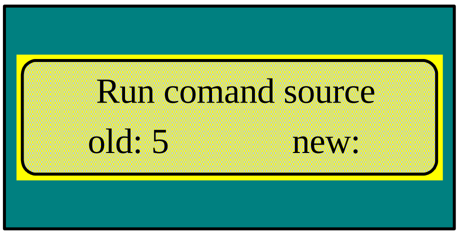
If the value is not 0, set it as 0 first and press  + 

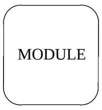
to confirm.



Set the value as 5, press  + 

to confirm.



Press 

to return to the main menu, the door operator close the door in place and the moment is kept.

After the adjustment, the pulling protection vane bolt must be removed, see Fig. 3-5.

3.4.3 Senior adjustment steps:

3.4.3 1 Power-on of door operator

Switch on the transducer switch; if the power indicator light of the transducer is on, the power is normal.

3.4.3.2 Enter password

Operate following 3.4.2.2.

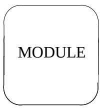
3.4.3.3 Orientation

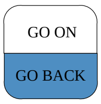
Under the condition of the operation mode being 0, pull the door to the middle of the door zone, enter 332 and then press ENTER, the door will move a little (a distance of about two centimeter), which means the orientation is finished. If the door does not move, change the door to another location manually to make the orientation again.

Enter menu 3313 to change the first parameter "rated power". The motor type is selectable. If Rated Power is set as 43.5, it is 2.3N·M motor; if it is set as 94.3, it's 5.0N·M motor.

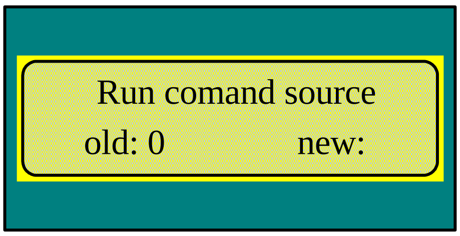
The motor parameter needs to be set again after replacing the motor or transducer.

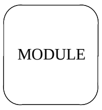
3.4.3.4 Setting direction of motor rotation

Press 

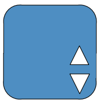
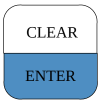
to return to the main menu, enter menu M 3311, then press 

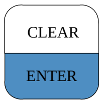
and select the following parameter. If it is not 0, set it as 0:



Pull the door to the middle, press 

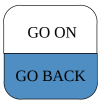
to return to the main menu, and enter menu M 311:

Press  + 

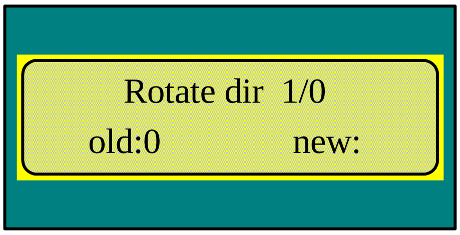
, then the door will operate towards one direction. Press 

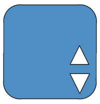
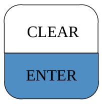
to stop it.

If the door operates towards the door opening direction, it's unnecessary to modify the motor rotation direction.

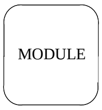
If the door operates towards the door closing direction, then enter menu M 3311, press 

to select the following parameter:



If the value is 0, modify it to 1; contrariwise modify it to 0. Press  + 

to confirm.

Press 

to return to the main menu.

3.4.3.5 Autotuning

Operate following 3.4.2.3.

3.4.3.6 Demonstration operation of door operator

Operate following 3.4.2.4.

3.4.3.7 Setting normal condition of door operator

Operate following 3.4.2.5.

Note: 1 All of the parameter including "Run command source" can only be modified when the parameter "Run command source" is 0.

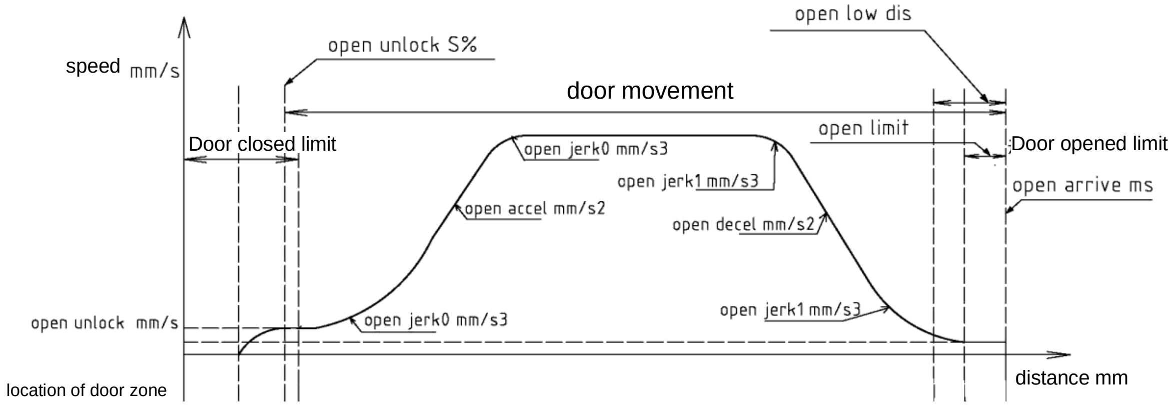
2 "Feedback mode" must be set as 0 during the adjustment, and it must be set as 1 after completing the adjustment.

3 "Run command source" can be set as 5 only if the car roof emergency button reset, the car roof maintenance mode and "Feedback mode" were all set as 1.

3.5 Opening-closing door curves

3.5.1 Opening door curve

See opening door curve in Fig. 3-6.

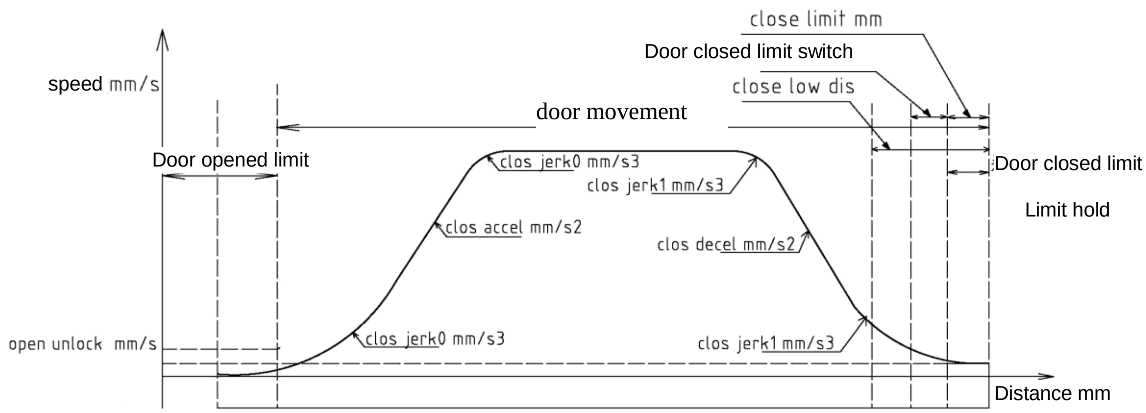


Parameter item	Menu	Function number, Location	Description
Open low dis mm	3311	4	Opened door running distance at low speed
Open unlock mm/s		5	Open, unlock speed
Open unlock S%		6	Open, unlock distance
Open accel mm/s2		7	Open, acceleration speed
Open jerk0 mm/s3		8	Open, acceleration fillet
Open decel mm/s2		9	Open, deceleration speed
Open jerk1 mm/s3		10	Open, deceleration fillet
Open limit	3315	15	Open, arrival position
Open arrive ms		17	Open, arrive arrival respond time

3—6 Drawing of opening door curve

3.5.2 Closing door curve

See closing door curve in Fig. 3-7.

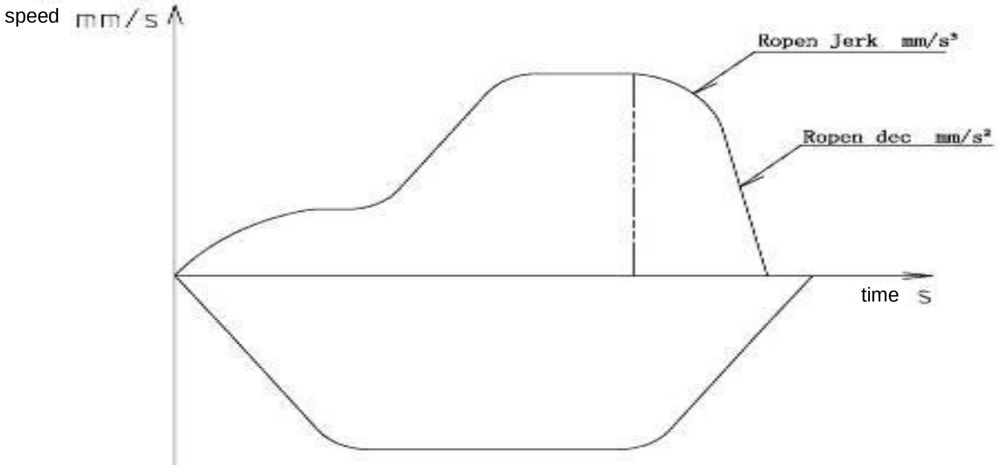


Parameter item	Menu	Function number, Location	Description
Close arrive mm/s	3311	11	Close, arrival speed
Close low dis mm		12	closed door running distance at low speed
Close accel mm/s2		13	close, acceleration speed
Close jerk0 mm/s3		14	Close, acceleration fillet
Close decel mm/s2		15	close, deceleration speed
Close jerk1 mm/s3		16	Close, deceleration fillet
Close unlock mm/s		17	Close Startup unlock speed
Close unlock S%		18	Close Startup unlock distance
Close limit mm	3315	19	Close, arrival position

Fig. 3-7 Drawing of closing door curve

3.5.3 Reopening door curve

See reopening door curve in Fig. 3-8.



Parameter item	Menu	Function number, Location	Description
Ropen Dec	3315	21	Reverse open door deceleration speed
Ropen Jerk		22	Reverse open door deceleration fillet

Fig. 3-8 Drawing of reopening door curve

3.6 Solution of common faults

The server can be used to check the fault code. Enter menu 321, and the current fault can be checked. Enter menu 322, and the fault history can be checked. Enter menu 324 and press ENTER, then the fault can be cleared.

3.6.1 Not-opening door

The control system sends the signals of opening and closing the door, but the door operator is not able to open or close the door.

Step1: enter 3312 to confirm that Feedback mode be set as 1.

Step1: enter 3311 to confirm that Run command source be set as 5.

Step3: inspect whether the wiring of RS 14 is connected reliably, the dial address of RS 14 is 53 and the setting of door opening-closing address for the control cabinet is correct.

3.6.2 Control system can not receive arrival signal

When the door reaches the closed position or opened position, the control cabinet can not receive the arrival signal of opening-closing from the door operator.

Step 1: when the door reaches the opened or closed position, observe the transducer for whether its correspondent limit light is on. If it is on, enter step 2; otherwise enter step 3.

Step 2: Pull off the plug-in P 1 to make the door operator operate automatically (.i.e. operating in the mode 1, 2, 5). Determine whether the connection and disconnection between P1-1 and P 1-2 runs normally as well as P1-1 and P 1-3. If they are normal, the transducer of the door operator operates well, and the fault occurs on the signal of the control system (the wiring of the control cabinet or the parameter setting). If they are not, the transducer of the door operator has been damaged.

Step 3: It's probably the too big door width of the autotuning that leads to the unfinished rear door width when the door is in opened limit. Check whether the value of DR half range (door width) in 3311 is consistent with the actual door width. The value is about half the actual door width + 25 mm.

3.6.3 Nonstop after reaching the opened limit

In this case, the harsh noise of belt slipping will come out generally, and there are two possibilities:

1, when operate manually (.i.e. Run command source = 0), the 311 operation command is sent, and the door will operate towards one direction. The door operator won't stop until the clear key is pressed. This is normal.

2, this phenomenon usually occurs during the autotuning. Check whether the belt is loose.

3, the location angle is not accurate, and it's necessary to relocate (pull the door to the middle).

3.6.4 Repeat of door opening-closing

Step 1: if the limit light is on when the door reaches the opened limit, the door operator has reached the limit. In this case, if the door operator closes without sake, you should check whether the control system has sent the signal of closing to the door operator.

Step 2: if the limit light is not on when the door reaches the opened limit, check whether the value of the door width is too big. Commonly double the door width is equal to the actual door width + 50mm.

Step 3: make sure that the closed limit input and output is constantly on when the door reaches the closed limit, and the closed limit input switch is effective.

3.6.5 Noise of door cutter

There is noise of the door cutter at the end of closing or at the beginning of opening: adjust the door cutter according to the mechanical adjustment instruction in the instruction.

3.6.6 Checking fault

Check the fault code (322) with the server. See the following table for fault related information:

Fault designation	Possible cause	Solution
BASE FAULT	Transducer has foreign objects dropping into it or transducer has been damaged	Clear away the foreign objects, and make sure whether the transducer is damaged.
	Any two of the three phases of the motor is short circuit	Check whether the three-phrase resistance of the motor is balance using multimeter. 2.3 NM motor is about 40 ohm, and 5.0 NM motor is about 30 ohm. If the three-phrase is not balance or the gap between the resistance value is too big, the motor has been damaged.
DC link OVT	Input voltage is too high	Check input power supply voltage
PVT lost	Encoder circuit of transducer is abnormal	Replace transducer
	Connection wire of encoder is disconnected or short circuit	Rearrange connection wire of encoder
	Hardware of encoder is damaged	Replace motor
MOTOR OVERLOAD	Something suffocates the process of opening and closing door	Check for foreign objects or mechanical blockage or too big door width
	Motor lacks phase during running of door operator	Check motor power line
	Wire disconnection of encoder	Rearrange connection wire of encoder
POWER LOST	Any one of the three phases of motor is short connected to ground	Measure the resistance of the three phrases of motor and the earth, and it should be infinite.
	Input voltage is too high	Measure input voltage, and it should be 220 V±20%

3.7 Transducer parameter sheet

Transducer parameter as shown in the following table

Tab. 3-3 Parameter sheet of door operator transducer

Function No.	Name	SVT display	Setting range	Default	Scope	Attribute
M311 (monitoring parameter group A)						
0	Software version	Easy-CON	As per actual condition			
1	Operation frequency	Freq out hz	0~50.00	※		
2	Motor rotation speed	Motor Speed RPM		※		
3	Rotor position	Rotor position	0~359.9	※		
4	Given velocity	Dictated V mm/s		※		
5	Given output voltage	Output voltage	0~900V	※		
6	Given torque current	Mtr trq PU	1.0—> rated torque	※		
7	Output current	Output current	0~999.9A	※		
8	Upper digits of run count	Run count(10000)		※		
9	Lower digits of run	Run Count1		※		

Function No.	Name	SVT display	Setting range	Default	Scope	Attribute
	count					
M312 (monitoring parameter group B)						
0	DC busbar voltage	DC link V	9999v	※		
1	ST code state	ST1 ST2 ST3	000000	※		
2	Encoder position	ENCODER POSITION	0~65536	※		
3	Current position	Door position	00000	※		
4	INPUT1	INPUT1		※		
5	INPUT2	INPUT2		※		
M313 (monitoring parameter group C)						
0	Open door input	Key open door False		※		
1	Close door input	Key close door False		※		
2	Stop	Stop press ENTER False		※		
M3311 (field adjustment parameter Field Adjust) EEPROM						
0	Obstruction contact memory function	Obistruction REM	0: disabled 1: enabled	0		invalid
1	Running direction reverse	Rotate dir 1/0	0,1	0		
2	RUN command source	RUN comd source	0~5 1、 Single running 2、 Continuous running 3、 DO/DC 4、ST code	0		
3	Door range	DR half range mm	0~65535	■		
4	Running distance at low speed of opened door arrival	Open low dis mm	0～100	5		
5	Open Startup unlock speed	Open unlock mm/s	5~100mm/s	45/60		
6	Open Startup unlock distance	Open unlock s ％	0~30.0%	3.0%/5%		
7	Open Acceleration speed	Open Accel mm/s2	10~2048mm/s/s	1000		
8	Open Acceleration fillet	Open Jerk0 mm/s3	10~2048mm/s/s/s	1000		
9	Open Deceleration speed	Open Decel mm/s2	10~2048mm/s/s	800		
10	Open Deceleration fillet	Open Jerk1 mm/s3	10~2048mm/s/s/s	800		
11	Close Arrival speed	Close arrive mm/s	5~100mm/s	15/50		
12	Running distance at low speed of closed door arrival	close_low_dis mm	5~100	15/50		
13	Close	Close Accel mm/s2	10~2048mm/s/s	800		

Function No.	Name	SVT display	Setting range	Default	Scope	Attribute
	Acceleration speed					
14	Close Acceleration fillet	Close Jerk0 mm/s3	10~2048mm/s/s/s	800		
15	Close Deceleration speed	Close Decel mm/s2	10~2048mm/s/s	600		
16	Close Deceleration fillet	Close Jerk1 mm/s3	10~2048mm/s/s/s	600		
17	Close Startup unlock speed	Close_unlock_mm/s	5~150mm/s	35		
18	Close Startup unlock distance	Close_unlock_s%	0~30.0%	0		
19	Password	Password	0~9999	8888		
M3312 (regulator parameter Regulator)						
0	Proportion gain 1 (high speed)	SpdP1 gain	0~10000	2500/2000		
1	Integration gain 1 (high speed)	SpdI1 gain	0~10000	1500/1000		
2	Proportion gain 2 (low speed)	Spdp2 gain	0~10000	2500/2000		
3	Integration gain 2 (low speed)	SpdI2 gain	0~10000	1500/1000		
4	PI transferring threshold value	SGP tran21h thr %	0~100	5		
5	PI transferring band width	SGP tran21 band%	0~100	5		
6	Speed feedback filter	Sfbk filter	0～66 (high and low)	33		
7	Given proportional part filter of the torque reference	Prop filter	0~3	0		
8	Motor overspeed	overspeed (pu)	1.00~2.00 1.0：Rated RPM	1.25		
9	Control method	Control methord	0:VF 1: Asynchronous motor closed loop 2: Synchronous motor	2		
10	Feedback mode	feedback mode	0：DO/DC 1：ST	1		
11	Current loop KP	CUR P gain	0~9999	512		
12	Current loop KI	CUR I gain	0~9999	150		
13	Electric torque limit (open door)	OP Drv Limit PU	0.00~2.50	2.50		
14	Braking torque limit (open door)	OP REG Limit PU	0.00~2.50	2.50		
15	Electric torque limit (close door)	CL Drv Limit PU	0.00~2.50	2.50		
16	Braking torque limit (close door)	CL REG Limit PU	0.00~2.50	2.50		
17	Motor overload	Mtr ovl i fac PU	0.1~2.0	1.2		

Function No.	Name	SVT display	Setting range	Default	Scope	Attribute
	protection factor					
18	PVT protection sensitivity factor	PVT Protect Gain	0~10 0：disable The larger the value is, the lower the sensitivity is.	03		
M3313 (motor parameter Motor Parameters)						
0	Motor power	Rated Power	0.1~999.9KW	43.5/94.3		
1	Number of poles	Number of poles	2~100	008		
2	Motor rated rotation speed	Rated RPM	1~9999	0180		
3	Motor rated frequency	Rated frq	1.00Hz~99.99Hz	12.00		
4	Motor rated voltage	Rated voltage	0~999V	100/125		
5	Rated current	Rated I A	1.0~999.9	000.8/1.2		
6	Roller diameter	Shv diam mm	10~10000mm	00045		
7	Deceleration ratio	Gear ratio	1.0~100.0	001.0		
8	Rope ratio	Rope ratio	1~6	1		
9	Magnetic pole original angle	Rotor pos offset	0~65535	■		
10	Stator resistor	RESIST S	0.000~9.999欧姆	7.730/2.790		
11	Stator inductance	INDUCT S	0.0~999.9mH	357.0/252.4		
12	Rotor resistor	RESIST ROTOR	0.000~9.999欧姆	5.230/1.820		
13	Rotor inductance	INDUCT R	0.0~999.9mH	357.0/252.4		
14	Mutual inductance	Mutual induct	0.0~999.9mH	325.0/240.6		
15	No-load current	No load current	0.0~999.9A	001.0/2.7		
M3314 (drive parameter Drive Scaling Parameters)						
0	Drive No.	Drive Size	0~100	000		
1	Rated voltage	Drive Rtd Volt (V)	0~1000	0220		
2	Rated current	Drive Rtd i RMS (A)	0.0~999.9	002.5		
3	Current adjusting coefficient	Inv i fscale (A)	0.000~2.000	1.00		
4	Voltage adjusting coefficient	Bus fscale (V)	0.000~2.000	1.050		
5	Current limit (overcurrent)	Inv i limit (A)	0.0~999.9	005.0		
6	Busbar overvoltage point	Bus ovt (PU)	0.00~2.00	1.40		
7	Busbar undervoltage point	DC link UV (PU)	0.00~1.00	0.60		
8	Input voltage coefficient	Line fscale (V)	0.00~2.00	1.00		
9	Brake voltage	Brake pick V (PU)	0.00~1.00 1.00:1000V	0.35	·	
10	Dead time	Inv dead time (us)	2~20	03		
11	PWM compensation	Inv PWM comp (us)	0.00~2.00	1.00		
12	Overshoot factor	Inv k mod (PU)	0~100	100		
13	Carrier frequency	Switch frq	0～12 KHz	12		

Function No.	Name	SVT display	Setting range	Default	Scope	Attribute
14	Running time (hour)	Power on Hour	0~65535H Need to be saved during power loss	※		*
15	Running time (minute)	Power on Minute	0~59MIN Need to be saved during power loss	※		*
M3315 (enhanced parameter Enhanced Parameters)						
0	Open/close operation hold time	Run hold time	0~99.9s 0：0: keep continuous operation Others: stop when the time is up	00.0		
1	Power on operation speed	Power on V mm/s	10~100mm/s	0050		
2	Door range detection speed	LEaRN V mm/s	10~100mm/s	0050		
3	Close arrival respond time	close arrive	200~3000ms	1000		
4	Open arrival respond time	open Arrive	500~3000ms Hold torque after arrival	800		
5	Arrival signal setting	Arr sw select	0: with open arrival and close arrival 1: without open arrival but with close arrival 2: without open arrival and close arrival	1	-	
6	Open Hold torque	Open hold torque	0~200.0%	100.0%		
7	Close Hold torque	Close hold torque	0~200.0%	60.0%		
8	Baffle detection identification time	Baffle timer	0~999ms 0：No function	200		
9	Baffle torque at high speed	Baffle torque Hi	0~200.0% Baffle torque during acceleration	140.0%		
10	Close baffle torque at middle speed	Baffle torque MID	0~200.0% Retarding torque for when ACC = 0	120.0%		
11	Baffle torque at low speed	Baffle torque Lo	0~200.0% Baffle torque during deceleration	80.0%		
12	RY1 function selection	RY1	0: open arrival signal (switch or pulse) 1: close arrival signal (switch or pulse) 2: fault output 3: baffle detection output 4: open door output 5: close door output 6: limited open arrival signal 7: limited close arrival	0		
13	RY2 function selection	RY2		1		
14	RY3 function selection	RY3		2		

Function No.	Name	SVT display	Setting range	Default	Scope	Attribute
			signal			
15	Open Arrival position	Open limit mm	0~1000mm	10		
16	Open Speed	Open speed	0~1000mm/s	0508		
17	Open Arrival speed	Open arrived V	5~100mm/s	020		
18	Door cutter selection	Door Coupler	0：Synchronous door cutter 1：Synchronous door cutter	0		
19	Close Arrival position	Close limit mm	0~1000mm	5		
20	Close Speed	Close speed	0~1000mm/s	508		
21	Reverse open door Deceleration speed	ROpen Dec	500~9999mm/s/s	5000		
22	Reverse open door Deceleration fillet	ROpen Jerk	500~9999mm/s/s/s	5000		
23	Demonstration door opened limit hold time	Demo open hold S	0~999.9 78s	003.0		
24	Demonstration door closed limit hold time	Demo open hold S	0~999.9s	003.0		
25	Manual acceleration	Man accel	10~2048mm/s2	0300		
26	Manual deceleration	Man decel	10~2048mm/s2	0500		
27	Manual speed	Man speed (mm/s)	0~999mm/s	0050		
28	Master and slave status setting	Master or Slave	0: Master status. Reopening door is possible 1：Slave status. Reopening door is impossible	0		
29	Empty	EMPTY	0	00000		
M3316 (VF parameter)						
0	VF mode	VF mode	0: Linear 1: Square	0		
1	Torque boost	TORQUE_BOOST	0~50.0%	20.0%		
2	Automatic torque compensation limit	TORQUE_CMPS_LIMIT	0~100％	000		
3	Baffle identification frequency ratio (at high speed)	Baffle_freq_Hi	0~100.0%	070.0		
4	Baffle identification frequency ratio (at low speed)	Baffle_freq_Lo	0~100.0%	050.0		

Function No.	Name	SVT display	Setting range	Default	Scope	Attribute
5	Baffle identification switch frequency	Baffle_freq_sw	0~99.99HZ	00.50		
6	Baffle identification time	Baffle_freq_tm	0~5000MS	0100		

1. Parameters marked with ※ can only be checked but not modified.
2. Parameters marked with ■ is obtained by autotuning, and can be modified by manual input.

4 Maintenance

4.1 Periodical maintenance of the door operator

During the elevator maintenance, the door rail should be cleaned and lubricated. Also, the wear of the door hanging board should be checked. If it exceeds 1.5 mm, adjustment or replacement should be made in time.

The tightness of the synchronous belt should be adjusted regularly. The requirement for the tightness is shown in Figure 4-1.

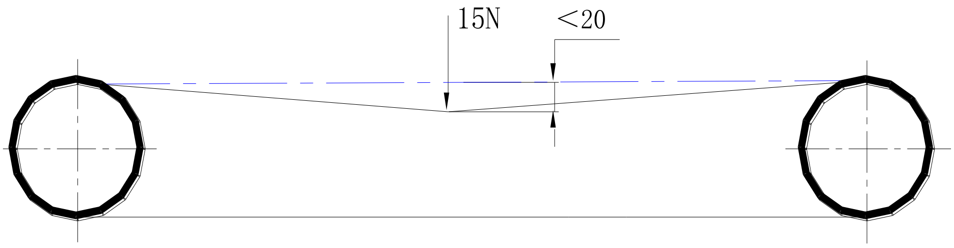


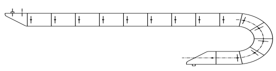
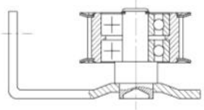
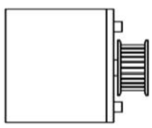
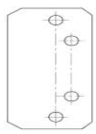
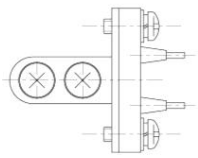
Figure 4-1 Requirement for tightness of synchronous belt

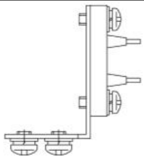
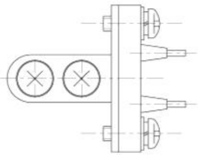
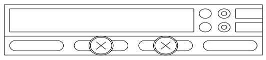
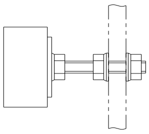
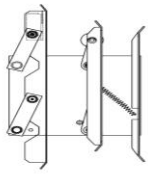
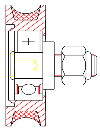
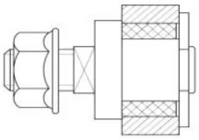
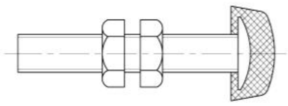
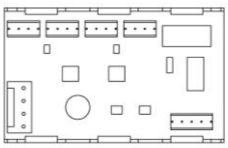
4.2 Daily maintenance of door operator

a) Check if the door could be opened and closed smoothly without any unusual sound.
b) Check regularly the tightness of all the fasteners.

If any problem is found during the above mentioned checking process, measures should be taken immediately to make sure that the door operator will operate normally.

5 Main parts list

Drawing no. of parts	Name	Cutline	Spare part (number in each)	Remark
XTA4215AAL	Synchronous belt with circular arc tooth		1	Easily damaged part
XTA4284AAR	Wire guard chain		2	Easily damaged part
XTA3052AAN	Tension assembly		1	
XTA2701ACE	motor		1	
XTA4386ABA	socket		Center-opening:3 Side-opening:4 Two-panel ceter-opening:4	
XTA3386AAY	Center plug assembly		Center-opening:3 Side-opening:4 Two-panel ceter-opening:4	

XTA3386ABP	Plug assembly		1 separately on the right and left	
XTA3386AAX	Plug assembly		1	
XTA3369AAE	Magnetic switch assembly		1 (used when safety edge is equipped)	常开
XTA3370AAA	Magnet assembly		1 (used when safety edge is equipped)	
XTA2703AAE	Pulling protection asynchronous door cutter		1	
XTA3035ADR	Hanging board wheel		4	Easily damaged part
XTB3117ABE	Hanging board adjusting roller		4	Easily damaged part
XTA3189AAG	Stopping part		4	
XTA3446AAK	transducer		1	
XTA4351ABJ	R 14 board		1	

Note: 1. if the customer needs the spare part of the door operator, please mark on the contract and provide relative list of spare part.(It's necessary to purchase the spare part separately)

2. The information in this document is subject to change without notice.